A FINITE CENTRAL SIMPLE ALGEBRA TENSORED WITH ITS OPPOSITE IS A MATRIX ALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the natural map is displayed with its formula and the two ingredients of the argument are named, so no step is left to the reader.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The tensor product with the opposite is central simple. Same chapter, Lemma lemma-tensor-central-simple. Supplies that the tensor product of the algebra with its opposite is simple, hence that the natural map into the endomorphism algebra is injective.
- The endomorphism algebra of a finite dimensional space. Same chapter, the identification of the endomorphism algebra of a finite dimensional vector space with a matrix algebra and its dimension. Supplies the dimension of the target of the natural map and its description as a matrix algebra.

Source reference. Original lemma 074G: sources/stacks-brauer/pinned-original-source.tex, source lines 555–555. Navigation pointer invoked at proof line 583; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let A be a finite central simple algebra over k. Then $A \otimes_k A^{op} \cong Mat(n \times n, k)$ where $n = [A : k]$.*

Proof. By Lemma 074G the algebra $A \otimes_k A^{op}$ is simple. Hence the map

$$A \otimes_k A^{op} \longrightarrow \mathrm{End}_k(A), \quad a \otimes a' \longmapsto (x \mapsto axa')$$

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is injective. Since both sides of the arrow have the same dimension we win. □